

Knowva ERD

Source: Notion DB 명세 v1.8 Mermaid ERD. 인증 수단별 독립 계정 정책과 DDL을 반영한 HTML 원본.

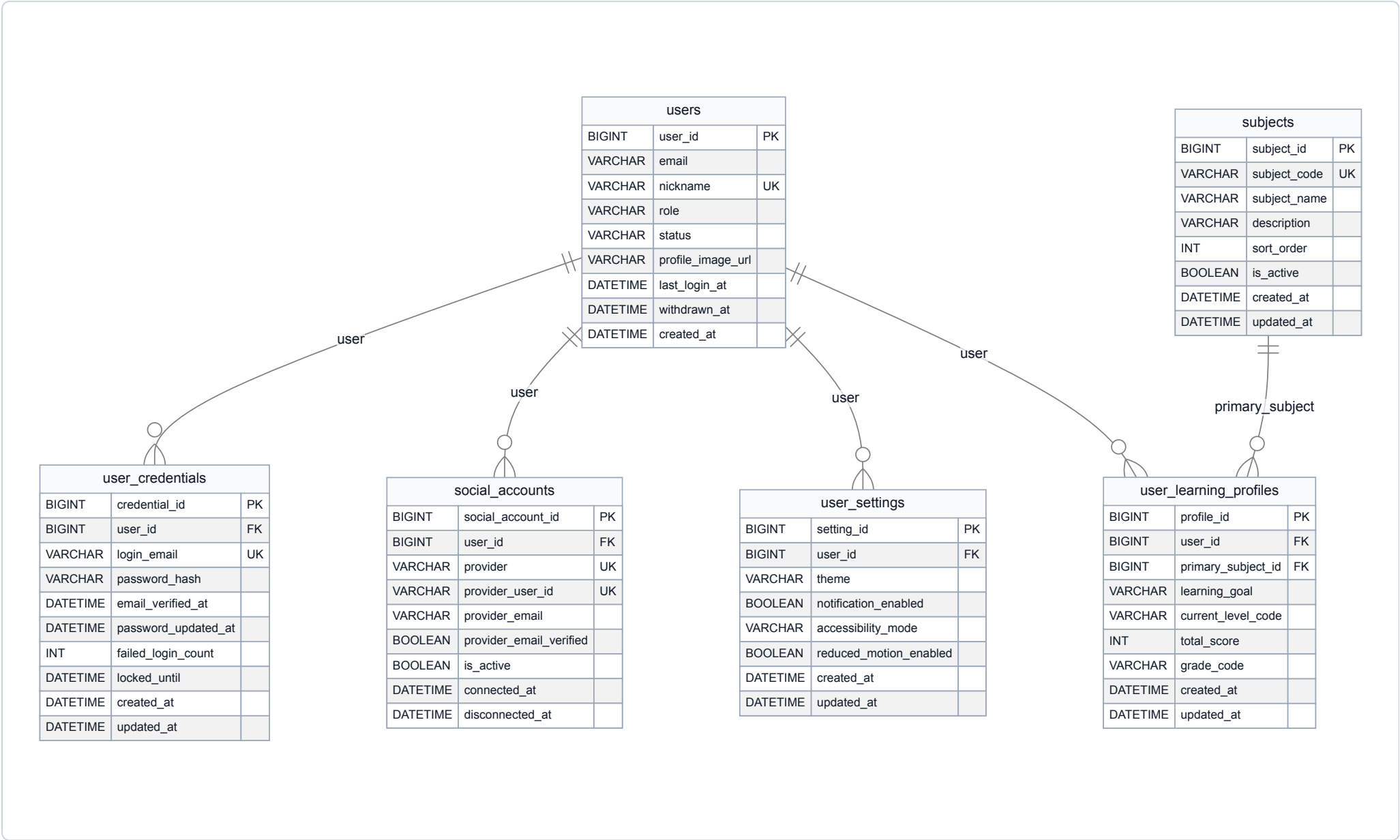
Version v1.8	Tables 40	Foreign Keys 71	DBMS MySQL 8
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Table	PK	FK Count
users	BIGINT user_id PK	0
subjects	BIGINT subject_id PK	0
curriculum_nodes	BIGINT node_id PK	2
payment_products	BIGINT product_id PK	0
user_credentials	BIGINT credential_id PK	1
social_accounts	BIGINT social_account_id PK	1
user_settings	BIGINT setting_id PK	1
user_learning_profiles	BIGINT profile_id PK	2
lessons	BIGINT lesson_id PK	2
lesson_bookmarks	BIGINT bookmark_id PK	2
level_test_questions	BIGINT question_id PK	1
level_test_choices	BIGINT choice_id PK	1
level_test_attempts	BIGINT attempt_id PK	2
level_test_answers	BIGINT answer_id PK	3
learning_progress	BIGINT progress_id PK	3
practice_set_attempts	BIGINT set_attempt_id PK	3
attendance_records	BIGINT attendance_id PK	2
practice_problems	BIGINT problem_id PK	3
problem_choices	BIGINT choice_id PK	1
practice_submissions	BIGINT submission_id PK	3
wrong_answers	BIGINT wrong_answer_id PK	4
score_events	BIGINT score_event_id PK	2
ranking_scores	BIGINT ranking_id PK	2
exam_sessions	BIGINT exam_id PK	2
user_level_unlocks	BIGINT unlock_id PK	3
ai_exam_problems	BIGINT ai_problem_id PK	1
exam_answers	BIGINT answer_id PK	2
ai_analysis_reports	BIGINT report_id PK	2

Table	PK	FK Count
ai_request_logs	BIGINT ai_request_log_id PK	0
community_posts	BIGINT post_id PK	2
post_attachments	BIGINT attachment_id PK	2
comments	BIGINT comment_id PK	3
post_likes	BIGINT like_id PK	2
post_scraps	BIGINT scrap_id PK	2
reports	BIGINT report_id PK	2
dummy_payments	BIGINT payment_id PK	2
premium_grants	BIGINT grant_id PK	2
notices	BIGINT notice_id PK	1
admin_operation_logs	BIGINT log_id PK	1
content_recommendations	BIGINT content_id PK	1

The diagram illustrates a complex network of interconnected nodes, each containing a table of data. The nodes are arranged in a hierarchical or clustered manner, with many lines connecting them, suggesting a complex system or database structure. The nodes are organized into several distinct groups, with some nodes acting as central hubs and others as peripheral elements. The connections between nodes are represented by lines, some of which are straight and others curved, indicating different types of relationships or data flows. The overall layout is dense and intricate, with a high degree of connectivity between the various data tables.

회원 / 인증 / 설정



회원 / 인증 / 설정 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
user_credentials	user_id	users	user_id
social_accounts	user_id	users	user_id
user_settings	user_id	users	user_id
user_learning_profiles	user_id	users	user_id
user_learning_profiles	primary_subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL		NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_credentials			
Column	Type	Key	NN
credential_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
login_email	VARCHAR(255) NOT NULL	UK	NN
password_hash	VARCHAR(255) NOT NULL		NN
email_verified_at	DATETIME NOT NULL		NN
password_updated_at	DATETIME NULL		NULL
failed_login_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
locked_until	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

social_accounts			
Column	Type	Key	NN
social_account_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
provider	VARCHAR(30) NOT NULL	UK	NN
provider_user_id	VARCHAR(191) NOT NULL	UK	NN
provider_email	VARCHAR(255) NULL		NULL
provider_email_verified	TINYINT(1) NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
connected_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
disconnected_at	DATETIME NULL		NULL

user_settings

Column	Type	Key	NN
setting_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
theme	VARCHAR(30) NOT NULL DEFAULT 'SYSTEM'		NN
notification_enabled	TINYINT(1) NOT NULL DEFAULT 1		NN
accessibility_mode	VARCHAR(30) NULL		NULL
reduced_motion_enabled	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

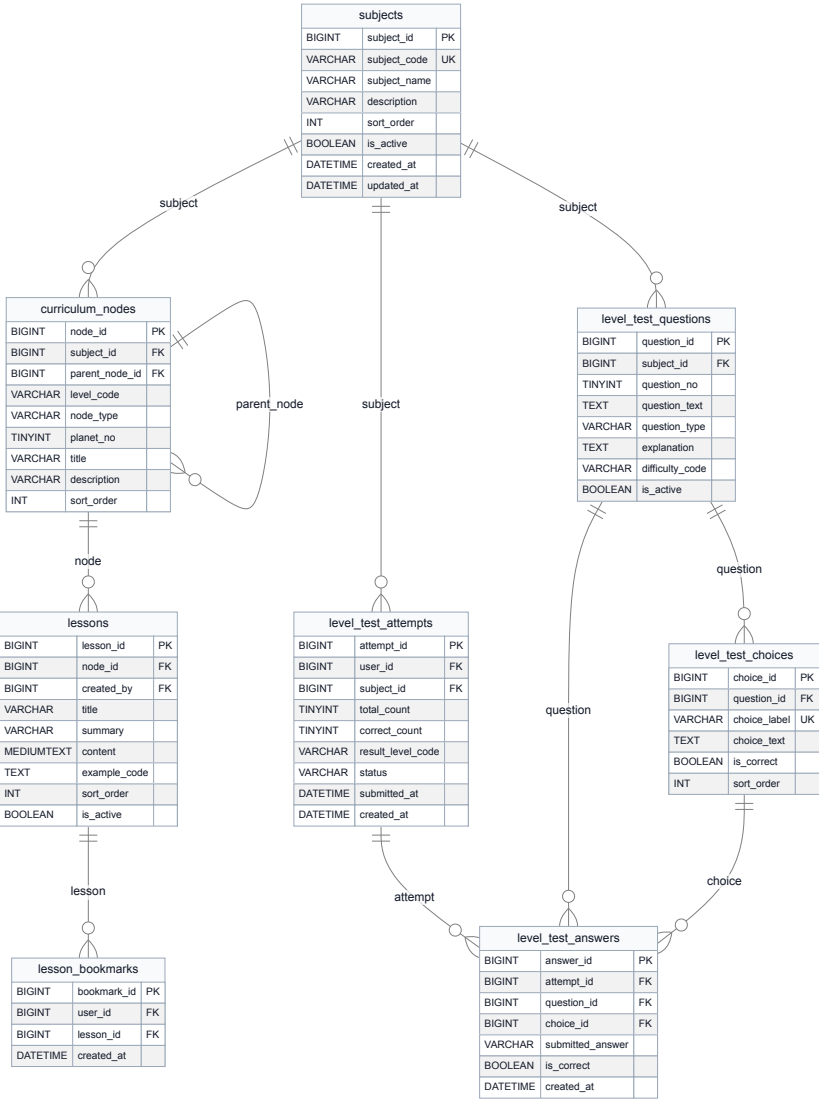
user_learning_profiles

Column	Type	Key	NN
profile_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
primary_subject_id	BIGINT UNSIGNED NULL	FK	NULL
learning_goal	VARCHAR(500) NULL		NULL
current_level_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
total_score	INT UNSIGNED NOT NULL DEFAULT 0		NN
grade_code	VARCHAR(30) NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

학습 Master / 레벨 테스트



학습 Master / 레벨 테스트 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
lessons	node_id	curriculum_nodes	node_id
lesson_bookmarks	lesson_id	lessons	lesson_id
level_test_questions	subject_id	subjects	subject_id
level_test_choices	question_id	level_test_questions	question_id
level_test_attempts	subject_id	subjects	subject_id
level_test_answers	attempt_id	level_test_attempts	attempt_id
level_test_answers	question_id	level_test_questions	question_id
level_test_answers	choice_id	level_test_choices	choice_id

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes			
Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lessons			
Column	Type	Key	NN
lesson_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(150) NOT NULL		NN
summary	VARCHAR(500) NULL		NULL
content	MEDIUMTEXT NOT NULL		NN
example_code	TEXT NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

lesson_bookmarks

Column	Type	Key	NN
bookmark_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_questions

Column	Type	Key	NN
question_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
question_no	TINYINT UNSIGNED NOT NULL		NN
question_text	TEXT NOT NULL		NN
question_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
explanation	TEXT NULL		NU LL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

level_test_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

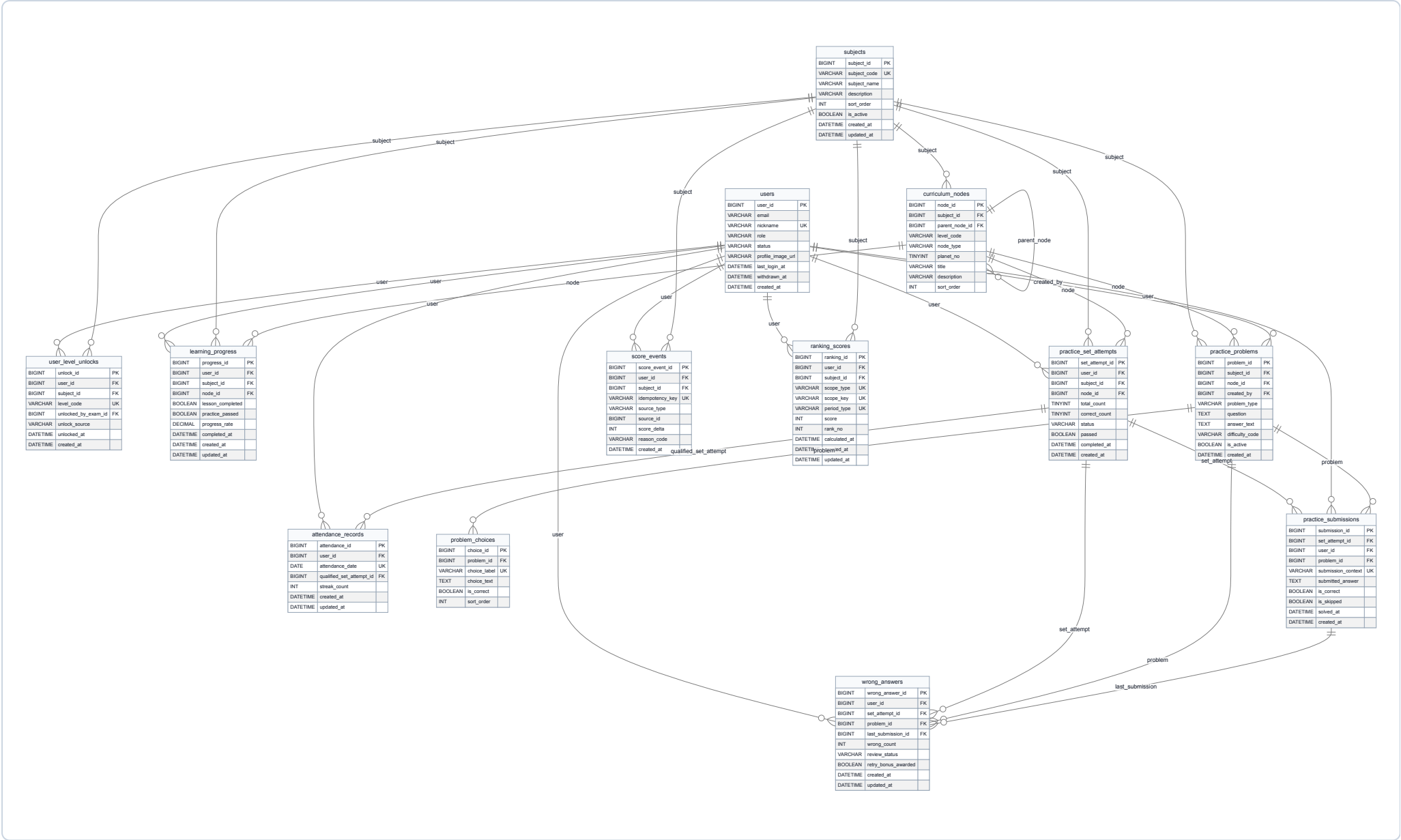
level_test_attempts

Column	Type	Key	NN
attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 8		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
result_level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'SUBMITTED'		NN
submitted_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

level_test_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
question_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_id	BIGINT UNSIGNED NULL	FK	NU LL
submitted_answer	VARCHAR(500) NULL		NU LL
is_correct	TINYINT(1) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

학습 기록 / 일반 문제 / 점수



학습 기록 / 일반 문제 / 점수 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
curriculum_nodes	subject_id	subjects	subject_id
curriculum_nodes	parent_node_id	curriculum_nodes	node_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
learning_progress	user_id	users	user_id
learning_progress	subject_id	subjects	subject_id
learning_progress	node_id	curriculum_nodes	node_id
attendance_records	user_id	users	user_id
attendance_records	qualified_set_attempt_id	practice_set_attempts	set_attempt_id
practice_set_attempts	user_id	users	user_id
practice_set_attempts	subject_id	subjects	subject_id
practice_set_attempts	node_id	curriculum_nodes	node_id
practice_problems	subject_id	subjects	subject_id
practice_problems	node_id	curriculum_nodes	node_id
practice_problems	created_by	users	user_id
problem_choices	problem_id	practice_problems	problem_id
practice_submissions	set_attempt_id	practice_set_attempts	set_attempt_id
practice_submissions	user_id	users	user_id
practice_submissions	problem_id	practice_problems	problem_id
wrong_answers	user_id	users	user_id
wrong_answers	set_attempt_id	practice_set_attempts	set_attempt_id
wrong_answers	problem_id	practice_problems	problem_id
wrong_answers	last_submission_id	practice_submissions	submission_id
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id
ranking_scores	user_id	users	user_id
ranking_scores	subject_id	subjects	subject_id

users

Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects

Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

curriculum_nodes

Column	Type	Key	NN
node_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_node_id	BIGINT UNSIGNED NULL	FK	NULL
level_code	VARCHAR(30) NOT NULL		NN
node_type	VARCHAR(30) NOT NULL DEFAULT 'PLANET'		NN
planet_no	TINYINT UNSIGNED NULL		NULL
title	VARCHAR(150) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
gate_condition	VARCHAR(255) NULL		NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NULL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

learning_progress

Column	Type	Key	NN
progress_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
lesson_completed	TINYINT(1) NOT NULL DEFAULT 0		NN
practice_passed	TINYINT(1) NOT NULL DEFAULT 0		NN
progress_rate	DECIMAL(5,2) NOT NULL DEFAULT 0.00		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

attendance_records

Column	Type	Key	NN
attendance_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
attendance_date	DATE NOT NULL	UK	NN
streak_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
qualified_set_attempt_id	BIGINT UNSIGNED NULL	FK UK	NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_set_attempts

Column	Type	Key	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
total_count	TINYINT UNSIGNED NOT NULL DEFAULT 10		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'IN_PROGRESS'		NN
passed	TINYINT(1) NOT NULL DEFAULT 0		NN
completed_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

practice_problems

Column	Type	Key	NN
problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
node_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_type	VARCHAR(30) NOT NULL DEFAULT 'MULTIPLE_CHOICE'		NN
question	TEXT NOT NULL		NN
answer_text	TEXT NULL		NULL
difficulty_code	VARCHAR(30) NOT NULL DEFAULT 'BRONZE'		NN
created_by	BIGINT UNSIGNED NULL	FK	NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

problem_choices

Column	Type	Key	NN
choice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
choice_label	VARCHAR(10) NOT NULL	UK	NN
choice_text	TEXT NOT NULL		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN

practice_submissions

Column	Type	Key	NN
submission_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
submission_context	VARCHAR(30) NOT NULL DEFAULT 'PRACTICE_SET'	UK	NN
submitted_answer	TEXT NULL		NULL
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
is_skipped	TINYINT(1) NOT NULL DEFAULT 0		NN
solved_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

wrong_answers

Column	Type	Key	NN
wrong_answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
set_attempt_id	BIGINT UNSIGNED NOT NULL	FK	NN
problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
last_submission_id	BIGINT UNSIGNED NULL	FK	NULL
wrong_count	INT UNSIGNED NOT NULL DEFAULT 1		NN
review_status	VARCHAR(30) NOT NULL DEFAULT 'OPEN'		NN
retry_bonus_awarded	TINYINT(1) NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

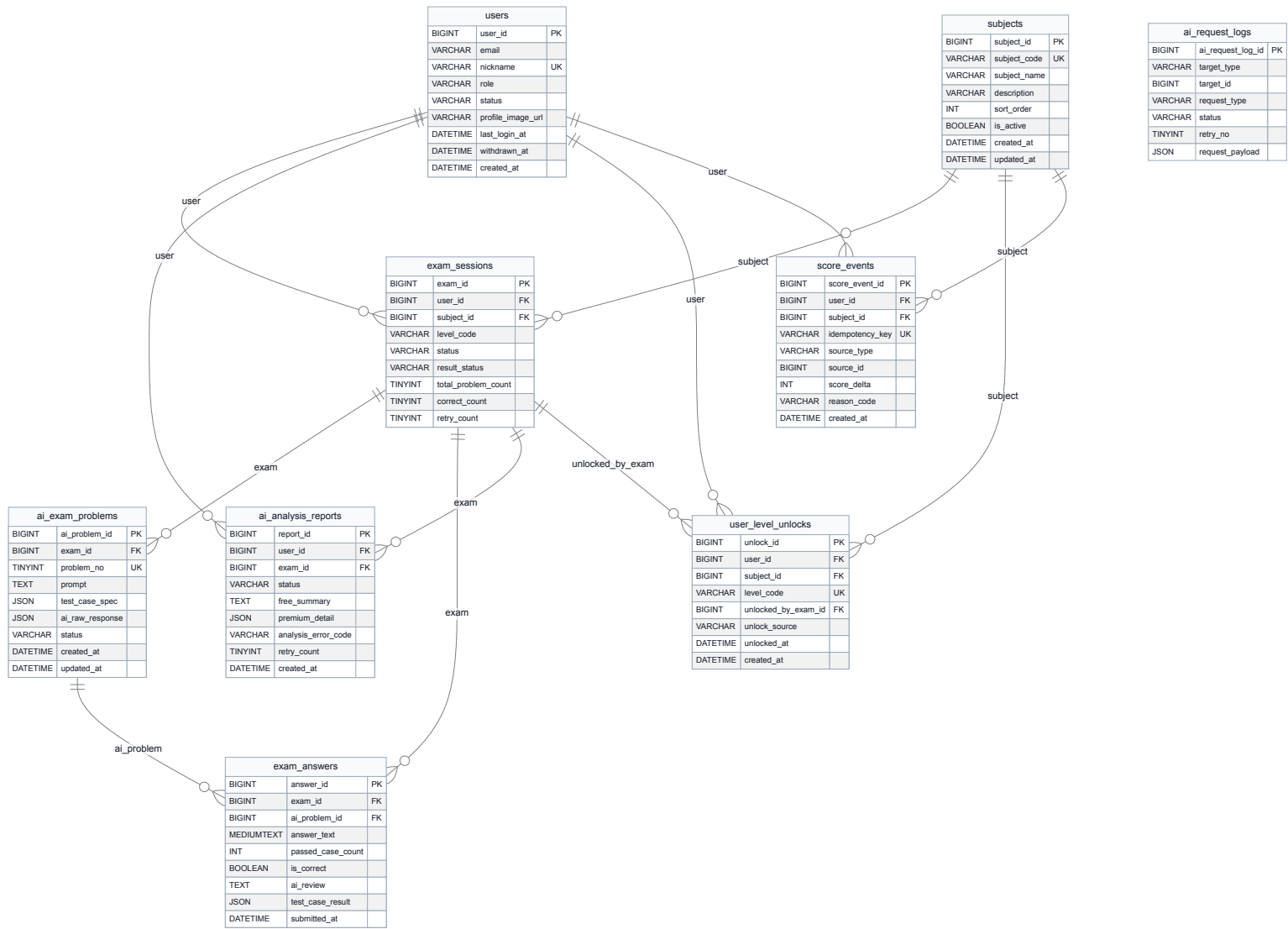
score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

ranking_scores

Column	Type	Key	NN
ranking_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NULL	FK	NU LL
scope_type	VARCHAR(30) NOT NULL	UK	NN
scope_key	VARCHAR(100) NOT NULL	UK	NN
period_type	VARCHAR(30) NOT NULL	UK	NN
period_key	VARCHAR(30) NOT NULL	UK	NN
score	INT UNSIGNED NOT NULL DEFAULT 0		NN
rank_no	INT UNSIGNED NULL		NU LL
calculated_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

AI 시험 / AI 분석



AI 시험 / AI 분석 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
exam_sessions	user_id	users	user_id
exam_sessions	subject_id	subjects	subject_id
ai_exam_problems	exam_id	exam_sessions	exam_id
exam_answers	exam_id	exam_sessions	exam_id
exam_answers	ai_problem_id	ai_exam_problems	ai_problem_id
ai_analysis_reports	user_id	users	user_id
ai_analysis_reports	exam_id	exam_sessions	exam_id
user_level_unlocks	user_id	users	user_id
user_level_unlocks	subject_id	subjects	subject_id
user_level_unlocks	unlocked_by_exam_id	exam_sessions	exam_id
score_events	user_id	users	user_id
score_events	subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

exam_sessions			
Column	Type	Key	NN
exam_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
level_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'CREATED'		NN
result_status	VARCHAR(30) NULL		NULL
total_problem_count	TINYINT UNSIGNED NOT NULL DEFAULT 3		NN
correct_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
started_at	DATETIME NULL		NULL
submitted_at	DATETIME NULL		NULL
graded_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

ai_exam_problems

Column	Type	Key	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
problem_no	TINYINT UNSIGNED NOT NULL	UK	NN
prompt	TEXT NOT NULL		NN
test_case_spec	JSON NULL		NU LL
ai_raw_response	JSON NULL		NU LL
status	VARCHAR(30) NOT NULL DEFAULT 'GENERATED'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

exam_answers

Column	Type	Key	NN
answer_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
ai_problem_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
answer_text	MEDIUMTEXT NOT NULL		NN
passed_case_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_correct	TINYINT(1) NOT NULL DEFAULT 0		NN
ai_review	TEXT NULL		NU LL
test_case_result	JSON NULL		NU LL
submitted_at	DATETIME NULL		NU LL
graded_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

ai_analysis_reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
exam_id	BIGINT UNSIGNED NOT NULL	FK	NN
status	VARCHAR(30) NOT NULL DEFAULT 'PENDING'		NN
free_summary	TEXT NULL		NU LL
premium_detail	JSON NULL		NU LL
analysis_error_code	VARCHAR(100) NULL		NU LL
retry_count	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

ai_request_logs

Column	Type	Key	NN
ai_request_log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
request_type	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL		NN
retry_no	TINYINT UNSIGNED NOT NULL DEFAULT 0		NN
request_payload	JSON NULL		NU LL
response_payload	JSON NULL		NU LL
error_code	VARCHAR(100) NULL		NU LL
error_message	VARCHAR(1000) NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

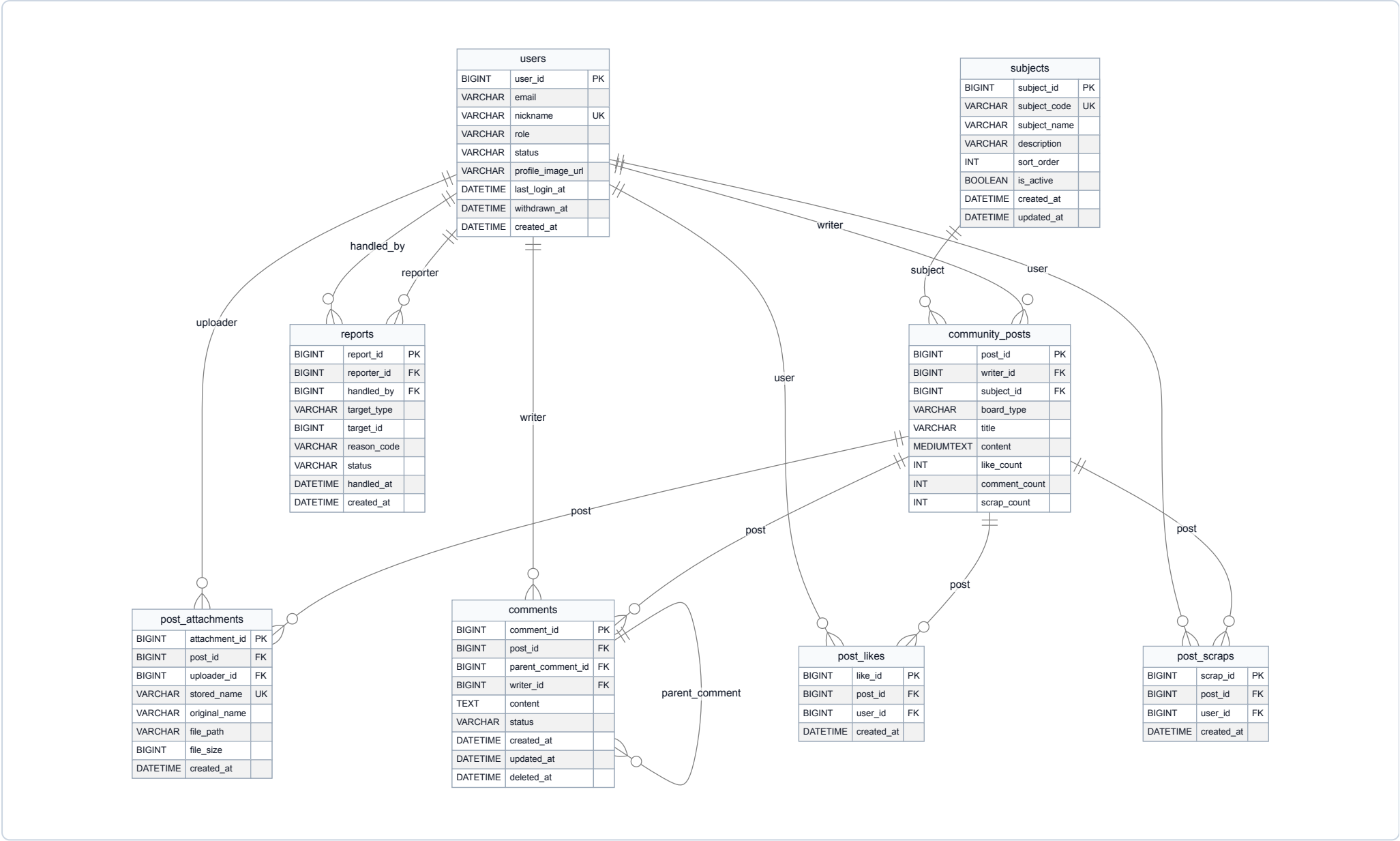
user_level_unlocks

Column	Type	Key	NN
unlock_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
level_code	VARCHAR(30) NOT NULL	UK	NN
unlock_source	VARCHAR(30) NOT NULL		NN
unlocked_by_exam_id	BIGINT UNSIGNED NULL	FK	NU LL
unlocked_at	DATETIME NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

score_events

Column	Type	Key	NN
score_event_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
source_type	VARCHAR(30) NOT NULL		NN
source_id	BIGINT UNSIGNED NOT NULL		NN
score_delta	INT NOT NULL		NN
reason_code	VARCHAR(30) NOT NULL		NN
idempotency_key	VARCHAR(191) NOT NULL	UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

커뮤니티



커뮤니티 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
community_posts	writer_id	users	user_id
community_posts	subject_id	subjects	subject_id
post_attachments	post_id	community_posts	post_id
post_attachments	uploader_id	users	user_id
comments	post_id	community_posts	post_id
comments	parent_comment_id	comments	comment_id
comments	writer_id	users	user_id
post_likes	post_id	community_posts	post_id
post_likes	user_id	users	user_id
post_scraps	post_id	community_posts	post_id
post_scraps	user_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

community_posts			
Column	Type	Key	NN
post_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
board_type	VARCHAR(30) NOT NULL		NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
like_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
comment_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
scrap_count	INT UNSIGNED NOT NULL DEFAULT 0		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL
deleted_at	DATETIME NULL		NULL

post_attachments

Column	Type	Key	NN
attachment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
uploader_id	BIGINT UNSIGNED NOT NULL	FK	NN
original_name	VARCHAR(255) NOT NULL		NN
stored_name	VARCHAR(255) NOT NULL	UK	NN
file_path	VARCHAR(500) NOT NULL		NN
file_size	BIGINT UNSIGNED NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

comments

Column	Type	Key	NN
comment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK	NN
parent_comment_id	BIGINT UNSIGNED NULL	FK	NU LL
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
content	TEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL
deleted_at	DATETIME NULL		NU LL

post_likes

Column	Type	Key	NN
like_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

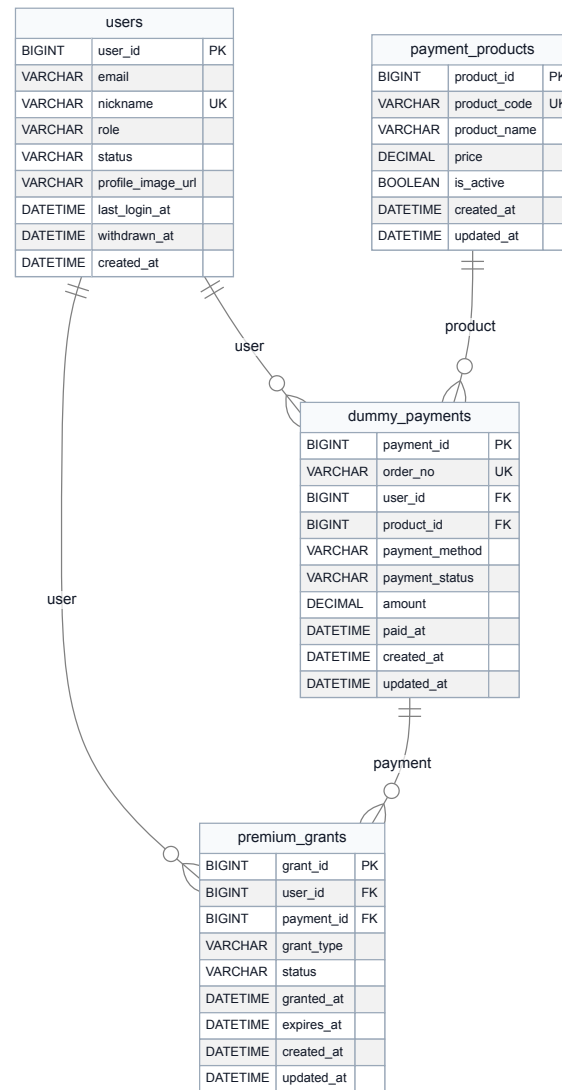
post_scraps

Column	Type	Key	NN
scrap_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
post_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK UK	NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

reports

Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NU LL
handled_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

결제 / Premium



결제 / Premium - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
dummy_payments	user_id	users	user_id
dummy_payments	product_id	payment_products	product_id
premium_grants	user_id	users	user_id
premium_grants	payment_id	dummy_payments	payment_id

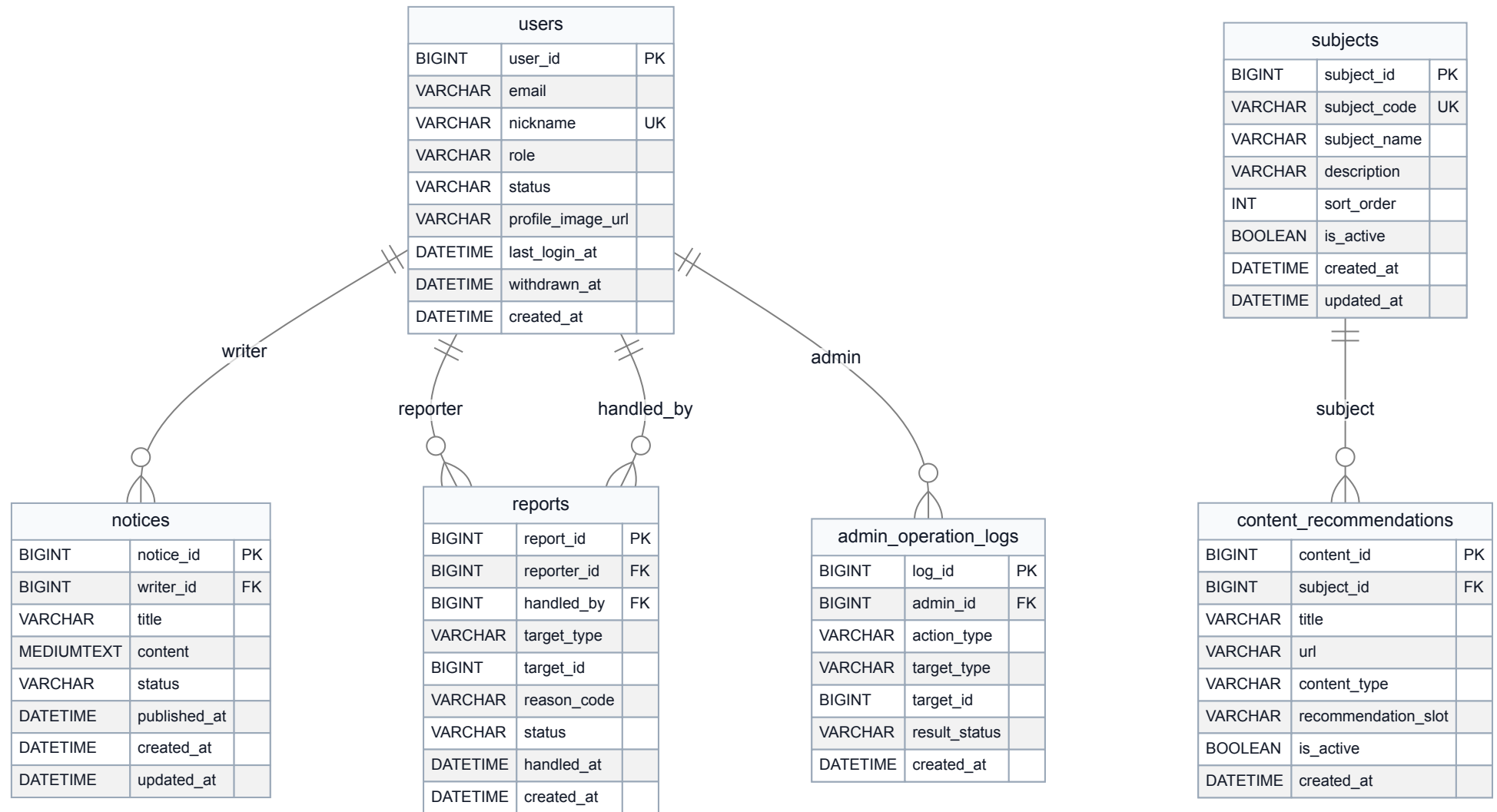
users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

payment_products			
Column	Type	Key	NN
product_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
product_code	VARCHAR(30) NOT NULL	UK	NN
product_name	VARCHAR(100) NOT NULL		NN
price	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

dummy_payments			
Column	Type	Key	NN
payment_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
order_no	VARCHAR(100) NOT NULL	UK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
product_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_method	VARCHAR(30) NOT NULL DEFAULT 'DUMMY'		NN
payment_status	VARCHAR(30) NOT NULL DEFAULT 'READY'		NN
amount	DECIMAL(10,2) NOT NULL DEFAULT 0.00		NN
paid_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

premium_grants			
Column	Type	Key	NN
grant_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
user_id	BIGINT UNSIGNED NOT NULL	FK	NN
payment_id	BIGINT UNSIGNED NULL	FK UK	NU LL
grant_type	VARCHAR(30) NOT NULL DEFAULT 'LIFETIME'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
granted_at	DATETIME NOT NULL		NN
expires_at	DATETIME NULL		NU LL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NU LL

관리자 / 공지 / 확장 콘텐츠



관리자 / 공지 / 확장 콘텐츠 - 관계 / 컬럼

Child Table	FK Column	Parent Table	Parent PK
notices	writer_id	users	user_id
reports	reporter_id	users	user_id
reports	handled_by	users	user_id
admin_operation_logs	admin_id	users	user_id
content_recommendations	subject_id	subjects	subject_id

users			
Column	Type	Key	NN
user_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
email	VARCHAR(255) NOT NULL	UK	NN
nickname	VARCHAR(50) NOT NULL	UK	NN
role	VARCHAR(30) NOT NULL DEFAULT 'ROLE_USER'		NN
status	VARCHAR(30) NOT NULL DEFAULT 'ACTIVE'		NN
profile_image_url	VARCHAR(500) NULL		NULL
last_login_at	DATETIME NULL		NULL
withdrawn_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

subjects			
Column	Type	Key	NN
subject_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_code	VARCHAR(30) NOT NULL	UK	NN
subject_name	VARCHAR(100) NOT NULL		NN
description	VARCHAR(500) NULL		NULL
sort_order	INT UNSIGNED NOT NULL DEFAULT 0		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

notices			
Column	Type	Key	NN
notice_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
writer_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
content	MEDIUMTEXT NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'DRAFT'		NN
published_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

reports			
Column	Type	Key	NN
report_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NOT NULL		NN
reporter_id	BIGINT UNSIGNED NOT NULL	FK	NN
reason_code	VARCHAR(30) NOT NULL		NN
status	VARCHAR(30) NOT NULL DEFAULT 'RECEIVED'		NN
handled_by	BIGINT UNSIGNED NULL	FK	NULL
handled_at	DATETIME NULL		NULL
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

admin_operation_logs			
Column	Type	Key	NN
log_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
admin_id	BIGINT UNSIGNED NOT NULL	FK	NN
action_type	VARCHAR(50) NOT NULL		NN
target_type	VARCHAR(30) NOT NULL		NN
target_id	BIGINT UNSIGNED NULL		NULL
result_status	VARCHAR(30) NOT NULL		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN

content_recommendations			
Column	Type	Key	NN
content_id	BIGINT UNSIGNED NOT NULL AUTO_INCREMENT	PK	NN
subject_id	BIGINT UNSIGNED NOT NULL	FK	NN
title	VARCHAR(200) NOT NULL		NN
url	VARCHAR(500) NOT NULL		NN
content_type	VARCHAR(30) NOT NULL		NN
recommendation_slot	VARCHAR(30) NOT NULL		NN
is_active	TINYINT(1) NOT NULL DEFAULT 1		NN
created_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP		NN
updated_at	DATETIME NULL DEFAULT NULL		NULL

v1.8 변경 요약

인증 수단별 독립 계정 정책

- `users.email`: UNIQUE 제거. 이메일은 연락처/표시 정보이며 계정 병합 기준으로 사용하지 않음.
- `user_credentials.login_email`: 이메일/비밀번호 로그인 식별자. 이메일 가입은 인증 완료 후 별도 계정으로 생성.
- `social_accounts.provider + provider_user_id`: Google/GitHub 소셜 로그인 식별자. `provider` 이메일이 기존 회원과 같아도 새 `users` 계정 생성.
- `provider`: google, github 소문자 값으로 통일.

v1.5 변경 요약

v1.4 AI 정책 반영

- `ai_exam_problems.test_case_spec`: ChatGPT API가 생성/보조한 테스트케이스 후보 저장
- `exam_answers.passed_case_count`, `test_case_result`: 정답/오답 판정은 테스트케이스 실행 결과로 처리
- `exam_answers.ai_review`: AI 해설/코드 리뷰 저장
- `ai_request_logs`: ChatGPT API 요청/응답/실패/재시도 `payload` 기록

구분	반영 내용
일반 문제 세트	<code>practice_set_attempts.subject_id</code> 추가. 문제 세트 <code>attempt</code> 에서 과목을 직접 추적.
오답 복습	<code>wrong_answers.set_attempt_id</code> 추가. 오답이 발생한 문제 세트 <code>attempt</code> 를 직접 추적.
로그인/권한	관리자는 사용자와 같은 로그인 화면을 사용하고, 로그인 후 <code>role</code> 에 따라 학습 또는 관리자 메인으로 이동.
AI Provider	로컬 LLM 미사용. ChatGPT API 호출/응답/실패/재시도 기록을 <code>ai_request_logs</code> 에 저장.
AI 코딩테스트	AI는 문제 생성, 테스트케이스 생성 보조, 해설 생성, 코드 리뷰 담당. 정답/오답은 <code>testcase</code> 실행 결과로 판정.
DDL 실행	전체 실행 시 <code>elearning schema</code> 의 <code>Knowva table</code> 을 <code>DROP TABLE IF EXISTS</code> 후 다시 생성.